

Bloomie

Problem

- Plants are sensitive to environmental changes which complicates teaching children about care
- Real plants may wither from neglect, while apps lack hands-on interaction for plant sensitivity

Method

- Created 3D models in Onshape for components including pond, electronics housing, and petal mechanism
- Integrate environmental sensors for light (photoresistor), heat (DHT11), water, and touch to read variables continuously
- Wire ESP32 microcontroller as brain to process sensor data, control stepper motor for petal movement, and update OLED display

Result

- Thinned walls in 3d model decreased material usage by 30%, lowering overall prototype weight from 450g to 315g
- Integrated stepper motor achieved 95% reliability in petal furling/unfurling based on threshold triggers from 4 sensors

